

A RESEARCH + DESIGN PROJECT • 2026

MYCELIUM

The Whisper Network an underground web of feeling

Whisper an anonymous sentence. An AI reads the feeling inside it and grows a small pixel creature – it drifts into a shared, living colony.
More connected than ever. Still disconnected.



React • canvas 2D • Claude • Edge • Redis
yueyao.design



gentle pip



misty bell



quiet wisp



marble mochi



sleepy pudding



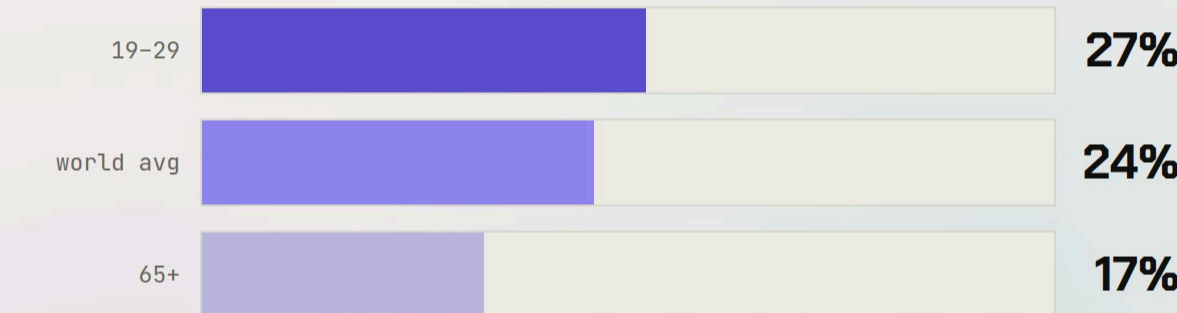
ember drift

01 • CASE STUDY

Background

Study abroad, relocation, a first job – transitions rarely end relationships; they cut the everyday synchrony those relationships ran on. Loneliness is the gap between the connection you want and the connection you feel (U.S. Surgeon General, 2023).

WHO FEELS IT MOST – BY AGE



~50%

of U.S. adults report loneliness

–70%

teen in-person time vs. 20 yrs ago

+29%

early-death risk ≈ 15 cigs/day

Meta-Gallup 2023 (142 countries) • U.S. Surgeon General 2023 • Pew • World Happiness Report 2025 • Baek et al., Psych. Science 2023

02 • CASE STUDY

Problem

Platforms make expression and contact visible, but not the perception of **collective emotional connection**.



“Very / fairly lonely” – Meta-Gallup 2023. *Europe: EU JRC 2022, stricter measure. China not among the 142 countries polled. What is missing is **perceivable emotional co-presence** – a different model of representation, not another feed.

03 · CASE STUDY

Target Users

18-30 young adults in transition – defined by life stage, not pathology.

■ In transition

study abroad · relocation · first job – students, migrants, graduates, remote workers.

■ Renegotiating relationships

across distance, time zones, language, identity, shifting roles.

■ Not a clinical group

life stage, not diagnosis – never framed as a disorder to be treated.

Missing: **perceivable emotional co-presence** – not another social feed.



fuzzy bundle

04 · CASE STUDY

Research Questions

How might **anonymous emotional input** become a **perceivable, evolving collective emotional environment**?

RQ2

without public performance

expression that never becomes performance

RQ3

AI as rule generator

not a conversational companion

RQ4

emotion → morphology

states mapped into form, behaviour, relation

RQ5

time-based ecological states

loneliness, support, transformation over time



moony moth



little dumpling

05 · CASE STUDY

Literature

SUPPORTS

loneliness as perceived disconnection, not a lack of people

young adults’ high platform use & loneliness

lonely people process the world idiosyncratically

care, ethics, contestability in sensitive design

humane data representation over dashboards

affect spreads through networks – cited critically

Cited as grounding, not as measured claims.



sunny sprite

SOURCE

Surgeon General (2023)

Pew · WHR (2025)

Baek et al. (2023), Psych. Science

Tad Hirsch

Giorgia Lupi – Data Humanism

Kramer et al.



misty bell

06 · CASE STUDY

Design Opportunity

From a platform logic to an ecological one.

FROM

direct communication

posting & replying

emotion as a label

AI as response generator

TO

indirect emotional co-presence

observing & inhabiting a shared field

emotion as form, behaviour & relation

AI as ecological rule generator

07 · DESIGN PRINCIPLES

■ Don’t diagnose, don’t treat

no scores, no therapy claims.

■ Emotion through form

indirect, slow, observable.

■ Care emerges from relations

rules, not a consolation button.

■ No companion, no performance

not a friend or chatbot.

■ Loneliness as a state

can change; not a label.

■ Keep ambiguity, show failure

uncertainty stays discussable.

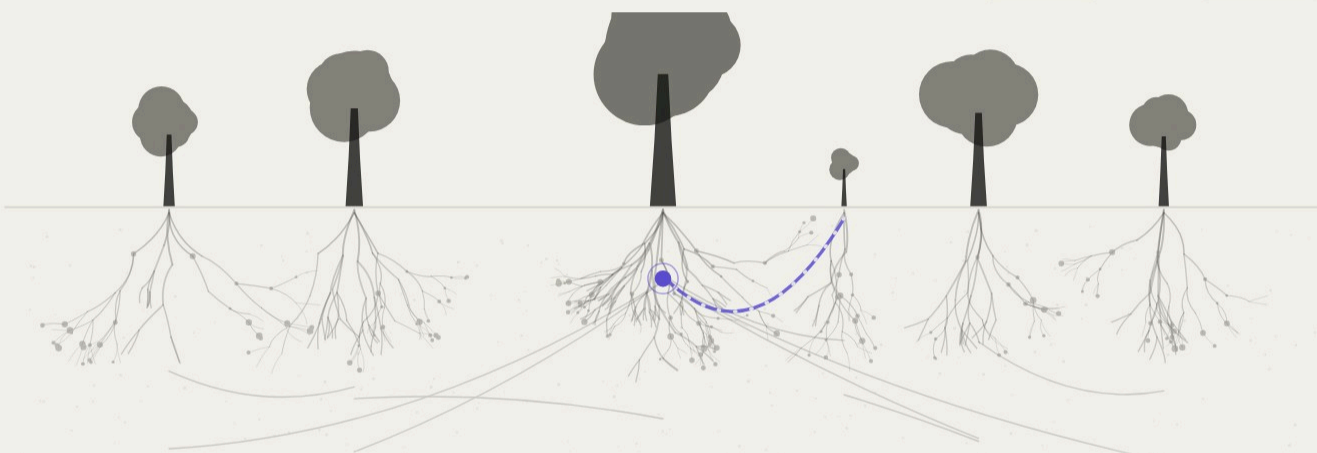


sunny crumb

08 • CASE STUDY

Biological Inspiration

Mycelial networks as a **metaphor** for indirect connection – a source of language, not a literal model; the biology is never offered as scientific proof.



BIOLOGY (AS INSPIRATION)

- underground network
- growth & decay
- mother trees supporting seedlings
- nutrient exchange

INTERACTION BEHAVIOUR

- indirect connection across a shared field
- entities emerge, fade, change over time
- mature entities support isolated ones
- relation leaves traces between bodies

The real wood-wide web: [Richard et al., Nature 388 \(1997\)](#) • [Klironomos et al., Science 352 \(2016\)](#) • [Beiler et al., New Phytologist 185 \(2010\)](#)

sunny sprite

honey bean

sleepy pudding

quiet wisp

velvet sprite

marble mochi

gentle pip

09 • CASE STUDY

The Role of AI

AI is framed as an **ecological rule generator** – not a chatbot, therapist, or emotional authority.

A disclosure becomes ecological parameters; the model sets the conditions for something to grow – it never writes back a comforting reply.

■ Interpretation, not truth

a reading that may be wrong – never a verdict on how you feel.

■ Boundaries are explicit

input → interpretation → parameters → organism. Never a persona.

■ Uncertainty is visible

errors are part of the system's story, open to contestation.

10 • CASE STUDY

System Overview

01

Input

tree-hole text or voice, anonymous

02

AI interpretation

Claude Haiku → labels + parameters

03

Morphology

parameters → a generated body

04

Social simulation

forces, compatibility, connection

05

Ecological feedback

loneliness, support, transformation

06

Observation

heatmap • gallery • debug

11 • USER JOURNEY

01

Arrive

a quiet living stage

02

Notice

the tree-hole input

03

Disclose

one anonymous sentence

04

Wait

the system interprets

05

Emerge

a new entity is born

06

Respond

neighbours gaze, greet

07

Dwell

bonds form, fade, transform

08

Explore

heatmap • gallery • debug

Interaction Flow

The interaction that matters is not typing the sentence – it is watching the social field rearrange itself afterward.



The tree-hole input on the live field – everything after this sentence is a system response.

STEP	USER DOES	SYSTEM RESPONDS
enter	arrives on the stage	ambient field, living population
disclose	writes / speaks	listening / interpreting state
emerge	watches	a new entity is generated
greet	observes	neighbours gaze & approach
evolve	stays	the field reorganises over time



calm bell



sunny crumb



honey bean



mint pip

Emotion Pipeline

A translation system, not an emotional diagnosis.

INPUT

Text / voice

one anonymous sentence

MODEL

Claude Haiku

interprets the disclosure

READING

Labels + intensity + topic

18 emotion labels

PARAMS

Morphology

six parameters

OUTPUT

Entity

a unique emotional body

A failed reading is handled cleanly instead of hidden – and, as future work, opened up so it can be challenged.



honey bean



mint pip

Morphological Mapping

PARAMETER

density

agitation

tendrilsCount

glow

tintHue

particles

BECOMES

body density • opacity • scale

jitter • breathing rhythm

connective tendrils

glow intensity

spore colour

spore-emission frequency

FORM

central-axis symmetry + per-row envelope + edge dither

COLOUR

tintHue + intensity widen the palette

EXPRESSION

procedural face – the pupil glances, blinks

RHYTHM

agitation drives jitter & breathing

low

a murmur

mid

a plain sentence

high

a cry

Same family, three intensities – systematic, never fully deterministic.



quiet wisp



peach sprite

15 • CASE STUDY

Biological Archetypes

Six prototype species – a **generative grammar, not fixed emotion mascots**. Morphology and hybridization mutate them; the field reads as one system while every member stays distinct.



tender
● H 24°



calm
● H 205°



curious
● H 32°



dreamy
● H 268°



companion
● H 158°



lonely
● desat.



sleepy pudding



little dumpling



moony moth



misty bell



sunny crumb



sunny sprite

16 • CASE STUDY

Social Simulation

A bond is not drawn as a line – it shows up as **tile-swap dyeing**: the tendrils are made of colour, from the point of contact.



Contact-point dyeing
the connection, as colour – it releases when they part



Matter exchange
bonded partners ferry mosaic tiles back and forth

BOND LIFECYCLE

meet

bonded

part

cooldown

SHOWN AS

the dye sweeps in from the contact side

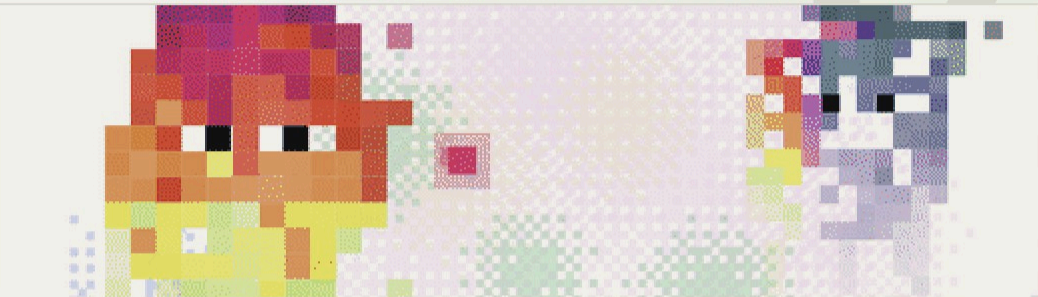
borrowed colour holds (≤45%, identity preserved)

the colour slowly releases

a pause before the pair can dye again

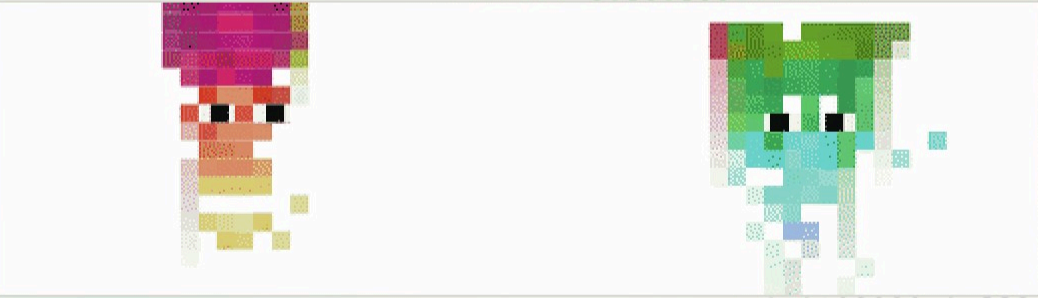
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Mother Tree



Care as an internal ecological mechanism
an entity that has stayed ~40s becomes a mother tree, reaching softer tendrils toward those left alone – nobody presses a button.

18 • HYBRIDIZATION



Encounter becomes memory & mutation
sustained contact: infection → transforming → hybrid – fifteen forms implemented.



sleepy pudding



little dumpling



moony moth



misty bell



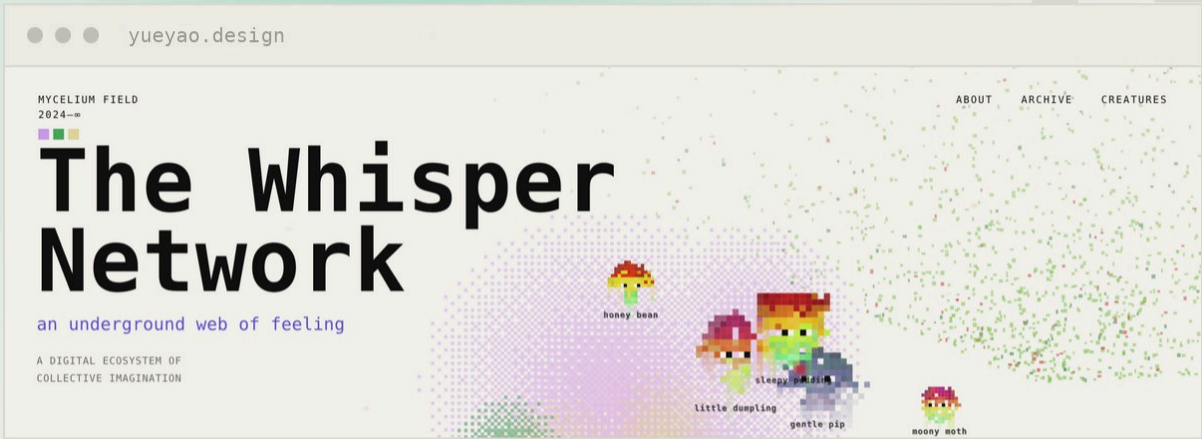
sunny crumb



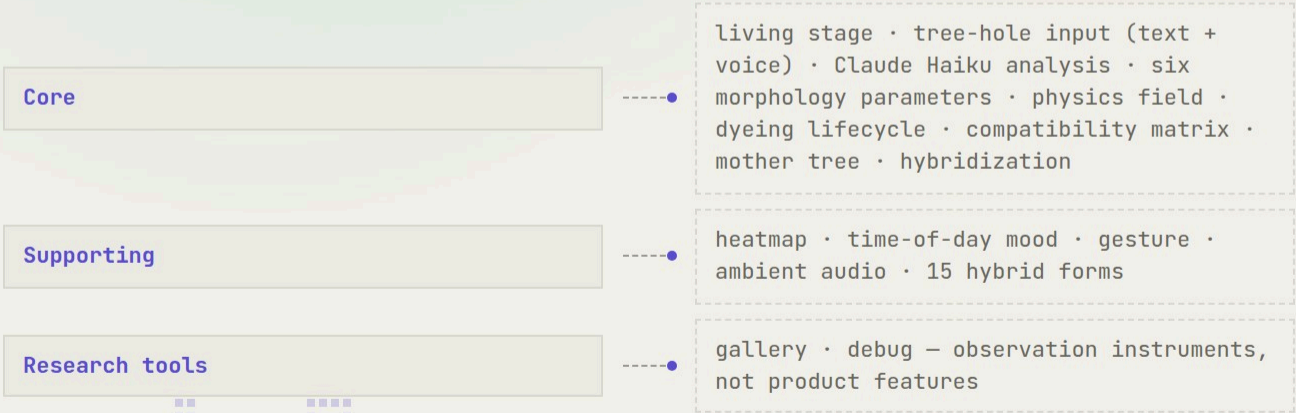
sunny sprite

19 • CASE STUDY

Prototype



ROLE



Every specimen in this document is rendered by the same deterministic engine — no hand-drawn art, no image model.



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Evaluation

Not whether it cures loneliness — whether it makes co-presence **perceivable, interpretable, discussable.**

METHODS	MEASURES	ETHICS
walkthrough • think-aloud	perceived co-presence	consent • privacy
interview	trust in AI mediation	emotional sensitivity
expert critique	comfort with anonymity	no therapy claims
diary • observation	concern about misreading	debriefing



Status: methods planned; the in-product survey (right) is live as one instrument. Findings reported qualitatively — no claims not gathered.



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Future Work

- AI contestability**
revise, reject, or reinterpret a reading.
- Long-term ecology**
traces across sessions and deployments.
- Community deployment**
situated studies with international students.
- Exhibition build-out**
the physical installation — sterile cultivation, sensor loop.
- Transparent morphology**
reveal how parameters shape form.
- Expanded evaluation**
walkthroughs & expert critique first.

The most important direction is not more features — it is making the system more transparent, easier to challenge, and grounded in an actual setting.



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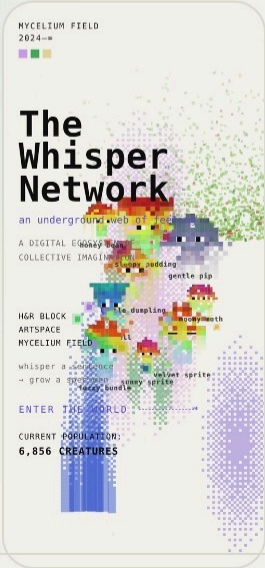
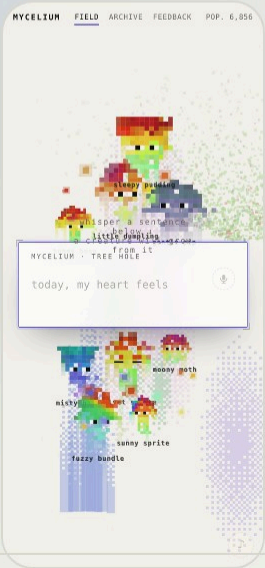
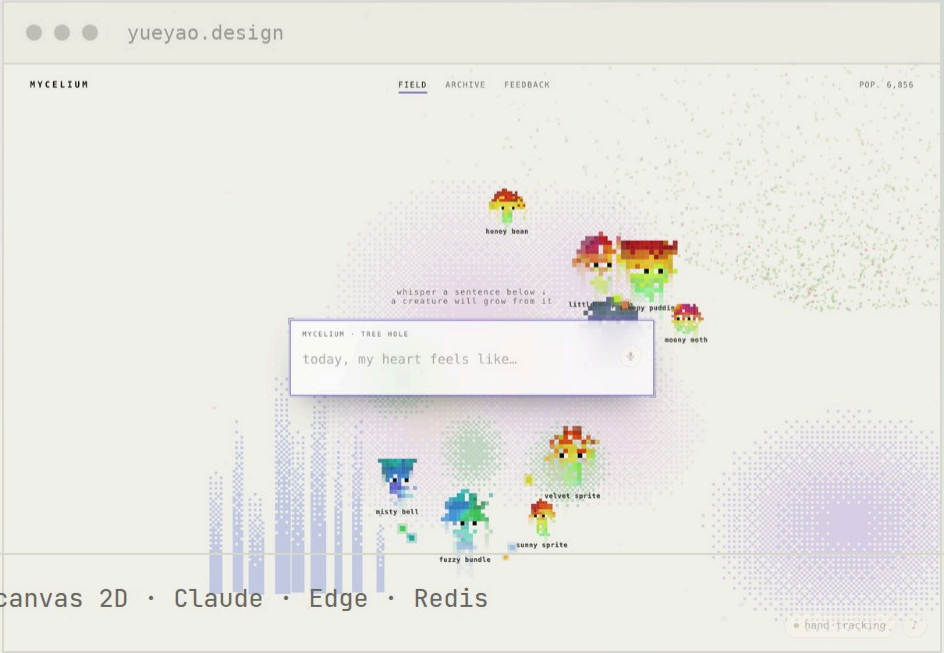
Reflection

A mature reflection makes the project **more credible, not weaker** – limitations become the next research questions: no therapeutic claims; AI can misread; the mycelial metaphor risks reading as cute; user research remains to be done. Naming these openly is part of the method.

The clearest research identity: **emotional visualization, AI mediation, social simulation, design ethics** – in the tradition of care-oriented design research (Hirsch).

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Final



The live prototype. Every creature is a real sentence someone whispered – held in a colony that keeps moving whether or not anyone is watching.

No accounts. No feed. No metrics.